

## HDOC 29

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**BATCH = 15**

### Algorithm

1. Start.
2. Display the menu:
  - Count even, odd, positive, negative, and zero elements.
  - Search for an element using linear search.
  - Merge two arrays.
  - Exit.
3. Read the user's choice.
4. Use `switch(choice)`:
  - **Choice 1:** Read the array size and elements. Traverse the array and count even, odd, positive, negative, and zero elements.
  - **Choice 2:** Read the array and the element to search. Traverse the array using linear search. Display its position if found; otherwise display "Element not found".
  - **Choice 3:** Read two arrays. Copy the elements of both arrays into a third array and display the merged array.
  - **Choice 4:** Exit.
  - Otherwise, display an invalid-choice message.
5. Repeat the menu until the user selects Exit.
6. Stop.

### Test-Case Table

| Test Case | Operation      | Input                         | Expected Output                                         |
|-----------|----------------|-------------------------------|---------------------------------------------------------|
| 1         | Count elements | 5 → 2 -3 0 4 -6               | Even = 4, Odd = 1, Positive = 2, Negative = 2, Zero = 1 |
| 2         | Count elements | 4 → 1 3 5 7                   | Even = 0, Odd = 4, Positive = 4, Negative = 0, Zero = 0 |
| 3         | Linear search  | 5 → 10 20 30 40 50, search 30 | Element found at position 3                             |
| 4         | Linear search  | 4 → 5 10 15 20, search        | Element not found                                       |
| 5         | Merge arrays   | 25<br>1 2 3 and 4 5           | Merged array: 1 2 3 4 5                                 |
| 6         | Invalid size   | Size = 0                      | Ask for a positive size again                           |
| 7         | Invalid        | Choice = 7                    | Invalid choice                                          |

|  |                |            |                    |
|--|----------------|------------|--------------------|
|  | menu<br>8 Exit | Choice = 4 | Program terminated |
|--|----------------|------------|--------------------|

## C Program

```
#include <stdio.h>
int main()
{
int choice;
do
{
printf("\n===== HDOC DAY-29 =====\n");
printf("1. Count even, odd, positive, negative and
zero elements\n");
printf("2. Search an element using linear search\n");
printf("3. Merge two arrays\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
{
int n, i;
int even = 0, odd = 0;
int positive = 0, negative = 0, zero = 0;
do
{
printf("Enter the number of elements: ");
scanf("%d", &n);
if (n <= 0)
printf("Size must be positive. Enter
again.\n");
} while (n <= 0);
int a[n];
printf("Enter %d elements:\n", n);
for (i = 0; i < n; i++)
{
scanf("%d", &a[i]);
}
}
```

```

for (i = 0; i < n; i++)
{
/* Count even and odd */
if (a[i] % 2 == 0)
even++;
else
odd++;
/* Count positive, negative and zero */
if (a[i] > 0)
positive++;
else if (a[i] < 0)
negative++;
else
zero++;
}
printf("\nEven elements = %d\n", even);
printf("Odd elements = %d\n", odd);
printf("Positive elements = %d\n", positive);
printf("Negative elements = %d\n", negative);
printf("Zero elements = %d\n", zero);
break;
}
case 2:
{
int n, i, key;
int position = -1;
do
{
printf("Enter the number of elements: ");
scanf("%d", &n);
if (n <= 0)
printf("Size must be positive. Enter
again.\n");
} while (n <= 0);
int a[n];

```

```
printf("Enter %d elements:\n", n);
for (i = 0; i < n; i++)
{
scanf("%d", &a[i]);
}
printf("Enter the element to search: ");
scanf("%d", &key);
/* Linear Search */
for (i = 0; i < n; i++)
{
if (a[i] == key)
{
position = i;
break;
}
}
{
\n",
if (position != -1)
printf("Element %d found at position %d.
key, position + 1);
}
else
{
printf("Element %d not found in the
array.\n", key);
}
break;
}
case 3:
{
int n1, n2, i;
do
{
printf("Enter size of first array: ");
scanf("%d", &n1);
if (n1 <= 0)
```

```

printf("Size must be positive. Enter
again.\n");
} while (n1 <= 0);
do
{
printf("Enter size of second array: ");
scanf("%d", &n2);
if (n2 <= 0)
printf("Size must be positive. Enter
again.\n");
} while (n2 <= 0);
int a[n1], b[n2], merged[n1 + n2];
printf("Enter %d elements of first array:\n",
n1);
for (i = 0; i < n1; i++)
{
}
scanf("%d", &a[i]);
printf("Enter %d elements of second array:
\n", n2);
for (i = 0; i < n2; i++)
scanf("%d", &b[i]);
/* Copy first array */
for (i = 0; i < n1; i++)
merged[i] = a[i];
/* Copy second array */
for (i = 0; i < n2; i++)
merged[n1 + i] = b[i];
{
}
{
}
{
}
}

```

```

printf("Merged array:\n");
for (i = 0; i < n1 + n2; i++)
{
printf("%d ", merged[i]);
}
printf("\n");
break;
}
case 4:
break;
printf("Program terminated.\n");
default:
printf("Invalid choice. Please enter a choice
from 1 to 4.\n");
}
} while (choice != 4);
return 0;
}

```

#### **Execution 1**

1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit

Enter your choice: 1

Enter the number of elements: 7

Enter 7 elements:

-1 -3 2 4 0 5 9

Even elements = 3

Odd elements = 4

Positive elements = 4

Negative elements = 2

Zero elements = 1

## Execution 2

```
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Enter the element to search: 30
Element 30 found at position 3.
```

## Execution 3

```
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter size of first array: 3
Enter size of second array: 4
Enter 3 elements of first array:
10 20 30
Enter 4 elements of second array:
40 50 60 70
Merged array:
10 20 30 40 50 60 70
```

#### Execution 4 - Exit

```
===== HDOC DAY-29 =====
```

1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit

```
Enter your choice: 4
```

```
Program terminated.
```